120 Days of Quantitative Finance

Week 8: Volatility Models & Returns Forecasting

Day 51: ARCH and GARCH Models

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1. Motivation

Classical return models assume constant variance:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2)$$

Empirical evidence shows volatility is time-varying:

$$Var(r_t|\mathcal{F}_{t-1}) = \sigma_t^2$$

This phenomenon is called conditional heteroskedasticity.

2. ARCH Model

ARCH(1) model:

$$r_t = \mu + \epsilon_t$$

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Conditions:

$$\alpha_0 > 0, \quad \alpha_1 \geq 0$$

Unconditional variance:

$$Var(r_t) = \frac{\alpha_0}{1 - \alpha_1}$$

provided:

$$\alpha_1 < 1$$

3. GARCH Model

GARCH(1,1):

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Stationarity condition:

$$\alpha_1 + \beta_1 < 1$$

4. Long-Run Variance

$$Var(r_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

5. Volatility Forecasting

One-step ahead forecast:

$$E(\sigma_{t+1}^2 | \mathcal{F}_t) = \alpha_0 + \alpha_1 \epsilon_t^2 + \beta_1 \sigma_t^2$$

Long-run forecast:

$$\lim_{h \to \infty} E(\sigma_{t+h}^2) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Conclusion

ARCH and GARCH models capture volatility clustering and persistence. They provide dynamic risk estimation and form the foundation of modern financial econometrics.